Electronics

What is electric Charge.

Scientist that comes to known that any Object that Can Push or pull each other without any Contact. Opposite Charge attract each other and same Charge Pushes away.

The substance Can easily transfer electric Charge is Conductor and the one Can't is the Insulator.

What is Voltage.

As we know that moving an electron in the same direction is called electricity. And the Voltage is used to measure that how much electron is running like the speed its travelling.

Potential Difference

What is Current:

The rate of the electron flow is called current.

What is Resistance

Resistance is used to resist the current flow. For example some materials are unable to pass the electron like Rubber, Silicon, wood etc....

Ohm's Law

Using the three property Current (I), Voltage (V) and Resistance (Ω) Ohm.

more voltage means more current. more current means more voltage which is the current and voltage are directly proportional to each other

current $\propto$ voltage

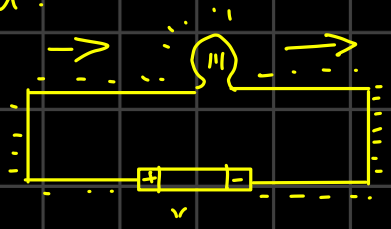
more Voltage more Resistance less Current

Voltage $\propto$ Resistance

Ohms law $= I = \frac{V}{R}$ or $V = I \times R$

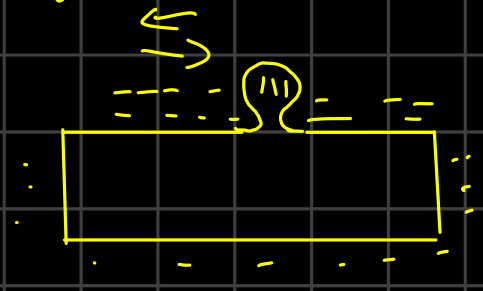
Direct Current (DC)

Moving of electron is electricity region From moving to one direction is called the direct current.

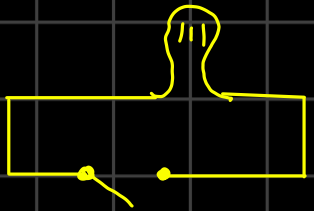


Alternative Current

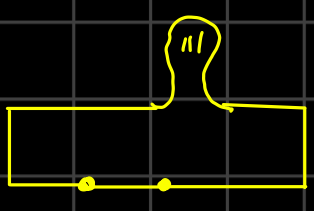
We know moving of electron is the electricity. And Instead of one direction we are moving the electron in the back and forth is called Alternative Current.



Open & Closed Circuit



open



closed



Switch

Open & Close

Parallel & Series Circuit



Series

Parallel

Parallel have seperate path so if one in trouble other still flow

Series will cut down both Power.

Electronics Components



Resistor

We know about the resistance that a physical component that used to make the electrical energy to Heat



Capasitor

Capasitor is used to store the electric energy with the Help of the Conductor Plate by seprating by dielect (Insulator) and release later

Diode

Diode is used to flow the current in one direction. ⟶|: Using this we can convert the AC to DC. Which is called Rectifier



Transistors

Transistor is the semiconductor device which used to control the electric flow by another electrical charge

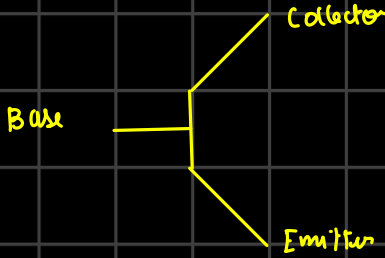
types of Transistor

BJT - Bipolar Junction Transistor has three terminal

collector - one path

emitter - other path

Base - Used to receive signal to control the flow



Example Another Current Give signal to Control the Circuit

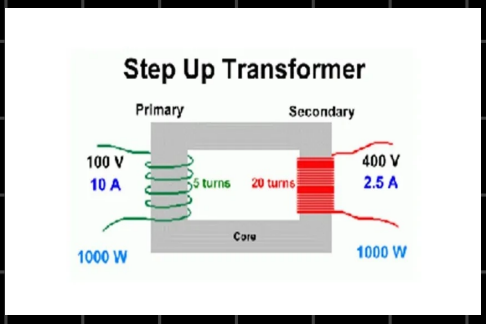
MOSFET - Same as that bcut the Voltage can be the Controller

Inductor

While the current flow the path have magnetic field and while we use coil we get strong magnetic field and using the we can store the charge in magnetic field and used to resist like resistor

Transformer

It is used to transfer the charge one coil to another using this we can convert the voltage higher to low or vise versa without wasting it with the help of magnetic energy



Voltage Regulator

The voltage regulator is used to stable the output voltage steadily. Example the input voltages is up & down but the output is steady "stable 5V"

Crystal Oscillator

The Crystal Oscillator is a Vibration device using that vibration it creates Frequency using that frequency we measure things

Integrated Circuit (ICs)

ICs is tiny electronic Component of many together in one small material. Used for space Occupation and depend on the work the different Component in it. Cost effective and Hard to design.